X-ray Crystal Structure of Anhydrous Chitosan at Atomic Resolution

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ABSTRACT:

We determined the crystal structure of anhydrous chitosan at atomic resolution, using X-ray fiber diffraction data extending to 1.17 Å resolution. The unit cell [a = 8.129(7) Å, b = 8.347(6) Å, c = 10.311(7) Å, space group $P2_12_12_1$] of anhydrous chitosan contains two chains having one glucosamine residue in the asymmetric unit with the primary hydroxyl group in the gt conformation, that could be directly located in the Fourier omit map. The molecular arrangement of chitosan is very similar to the corner chains of cellulose II implying similar intermolecular hydrogen bonding between O6 and the amine nitrogen atom, and an intramolecular bifurcated hydrogen bond from O3 to O5 and O6. In addition to the classical hydrogen bonds, all the aliphatic hydrogens were involved in one or two weak hydrogen bonds, mostly helping to stabilize cohesion between antiparallel chains. © *2016 Wiley Periodicals, Inc. Biopolymers 105: 361–368, 2016.*

Keywords: *chitosan; fiber X-ray diffraction; crystal structure*

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INTRODUCTION

Chitosan is a β-1,4-linked linear polymer of D-glucosamine, which is usually prepared by deacetylation of chitin, a homopolymer of *N*-acetyl glucosamine, most often using highly concentrated sodium hydroxide solution. Due to the regular and high density of amino groups, chitosan can tightly bind metal ions, and shows anti-microbial activities, probably due its polycationic nature. Therefore, chitosan is attracting considerable attention as a bio-based functional polymer.[1–4]

Chitosan can be obtained as fibrous crystalline material when converted from fibrous chitin using concentrated alkali solution. When the alkali-swollen chitosan is washed with water, a hydrated chitosan crystal is obtained.[5,6] The reported structure of hydrated chitosan has an orthorhombic unit cell with the space group $P2_12_12_1$ containing four molecular chains arranged in an antiparallel manner, and water molecules

Additional Supporting Information may be found in the online version of this article.

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having a certain disorder. In addition to the hydrated form of chitosan, a multitude of crystalline allomorphs can be obtained in complex with acids or metal ions. In these complexes, the chitosan molecule takes various helical structures.[7]

When the hydrated chitosan is heated in water, the water held in the hydrated crystal is expelled at around 200°C to yield anhydrous chitosan.[5,8] Anhydrous chitosan can also be obtained via complexation with monocarboxylic acids.[8,9] Hydrated chitosan is immersed in monocarboxylic acid to form carboxylic acid–chitosan salt. The acid and water molecules in the crystal are removed simultaneously when exposed to a high relative humidity. From solution state, short chain chitosan (DP 35) crystallizes into a lamella-shaped single crystal of anhydrous chitosan at 125°C.[10]

The molecular arrangement of the anhydrous chitosan crystal has been proposed based on diffraction studies using a limited number of diffraction intensities and samples prepared by different procedures. Mazeau et al.[11] refined the crystal structure using electron diffraction data of chitosan single crystals containing 23 independent base plane diffraction intensities ($h\ k\ 0$), and 24 $h\ k\ l\ (l \neq 0)$ intensities. Two sets of atomic coordinates determined by X-ray fiber diffraction have been reported using anhydrous chitosan prepared by hydrothermal treatment[12] and from chitosan/carboxylic acid complex.[13] These structures have in common an orthorhombic unit cell, which contains two chains with an antiparallel chain arrangement, but differ in details such as hydrogen bonding patterns due to slight differences in molecular disposition. The most recent structure determination by Okuyama et al.[13] was based on 41 diffraction intensity values under certain geometrical constraints and non-bonded interaction rationales.

During the last decade, a series of allomorphs of cellulose and chitin have been re-examined using high-resolution X-ray and neutron diffraction.[14–20] The acquisition of better data and refinement strategies erased some of the ambiguities of the trial-and-error type structure refinement. In the present study, we re-examined the structure of anhydrous chitosan using high-resolution X-ray data, and determined the structure without *a priori* constraints with higher precision and reliability, which allowed in-depth analysis of molecular interactions.

MATERIALS AND METHODS

Sample Preparation

The chitosan sample was prepared by deacetylation of oriented α-chitin obtained from a crab tendon. A crab, *Chionoecetes opilio*, was boiled in water for 15 min and the tendon was removed from its body. The tendon was purified by successive treatments in KOH aqueous solution and $NaClO_2$ aqueous solution, as described previously.[21] The purified tendon was then deacetylated by treatment with 50% sodium hydroxide solution at 120°C for 2 h. This treatment was repeated three times with water rinsing after each treatment. The obtained hydrated chitosan fiber was dried in ambient conditions under tensile stress to keep the axial orientation.

Anhydrous chitosan samples were prepared by hydrothermal treatment of the hydrated chitosan fiber. The sample was held between glass plates in order to preserve the orientation and treated in 0.1*N* NaOH at 240°C for 20 min using a pressure vessel. Then it was dried in ambient conditions while keeping the fiber length constant.



FIGURE 1 Fiber diffraction diagram of anhydrous chitosan. The blue dots denote to the positions of the calculated diffraction spots from the unit cell. The red dots are the positions of the diffraction spots that are extinct in the $P2_12_12_1$ symmetry.

X-ray Fiber Diffraction

The X-ray fiber diffraction pattern was recorded on a flat imaging plate (R-Axis V, Rigaku, Japan) at room temperature at the beam line BL40B2 at SPring-8 (Hyogo, Japan). The wavelength and beam size was 1.0 Å and 100 μm, respectively. The exposure time was 120 s and the sample-to-detector distance, about 17 cm, was calibrated using Si powder. The peak positions were measured using R-Axis display software (Rigaku, Japan). After indexing the reflections in the fiber diagrams, the unit-cell parameters were refined using the least-squares method. Polarization correction, background subtraction, and peak fitting were carried out as previously described.[19]

Unit Cell Determination

The unit cell parameters of anhydrous chitosan were determined in the previous study having an orthorhombic unit cell [$a = 8.129(7)$ Å, $b = 8.347(6)$ Å, $c = 10.311(7)$ Å][8] and the same diffraction pattern was used for structure refinement in this study. Faint intensities were observed on the meridian corresponding to 0 0 1 and 0 0 3, but we considered they were reflections from the surface chains, as in the case of chitin. The systematic absence of odd order reflections in the equator (Figure 1) suggests that the space group is $P2_12_12_1$.

Rigid Body Grid Search

At first, a rigid β-D-glucosamine residue without the pendant hydroxymethyl group and hydrogen atoms was used as the phasing model. The residue with standard $^4C_1$ conformation having the Cremer & Pople puckering parameter[22] $\theta = 3.5°$ was placed to form a chain around the 2_1 symmetry axis, parallel to the *c*-axis, to link itself by symmetry operation. Due to the symmetry requirement and the polymeric nature of chitosan, the degrees of freedom were limited to translation along the *c*-direction and rotation around the symmetry axis. The agreement with the intensity data was calculated by the agreement function;

$$E_I = 1 - \left(\frac{\boldsymbol{I}_{obs} \cdot \boldsymbol{I}_{calc}}{|\boldsymbol{I}_{obs}| \cdot |\boldsymbol{I}_{calc}|} \right)^2$$

where $\boldsymbol{I}_{obs}$ and $\boldsymbol{I}_{cal}$ are the arrays of the observed and calculated intensities, respectively.

The value E_I was very sensitive to rotation, having two minima, and the one corresponding to the final structure by Okuyama et al. was significantly lower. In both cases, the pyranose ring is slightly oblique to the *ac* plane, but with opposite sign. The two angles differ in the sense of overlap among the parallel chains. The translation along the *c*-axis dictates the mutual disposition among the antiparallel chains. Again, there were two minima corresponding to structures with the centers of the pyranose ring shifted with respect to its antiparallel neighbor, both by roughly half the length of the residue but in a different sense.

Refinement

The X-ray structure refinement was carried out using SHELX-97[23] as previously described.[17] The two different minima of the translations determined above were used as starting coordinates: the *z*-coordinate of C1 at 0.5 (Model 1) and at 0.7 (Model 2) in fractional coordinate.

First, the bond lengths and bond angles were restrained to the average values of the β-1,4-D-glucosamine dimer[24] using the DFIX and DANG options of SHELX-97 with default standard deviations. Initial refinements were carried out with the O6 atom removed from the phase models, and using the crude intensity list up to 1.5 Å resolution. The refinements gave the classic crystallographic *R*-factors of $R1 = 0.37$ for Model 1, and $R1 = 0.43$ for Model 2.

Both $F_o - F_c$ and $2mF_o - F_c$ (σ) Fourier omit maps were calculated and visualized using *Coot*[25] and Raster3D.[26] In both models, a clear density peak that could be assigned to the O6 atom was observed close to the *gt* position (Figure 2). The missing O6 was then introduced at the position of the density peaks, followed by refinement using the crude data up to 1.5 Å resolution, and this gave *R*1 factors of 0.30 for Model 1, and 0.40 for Model 2, for 98 $F_o > 4\sigma(F_o)$ intensities. On the basis of this result, Model 2 was rejected at this stage.

Further structure refinements were carried out with the regrouped data list up to 1.17 Å resolution. With the bond lengths and bond angles restrained and global thermal displacement factors, the *R*1 factor was 0.16 for 101 $F_o > 4\sigma(F_o)$ intensities. Incorporation of aliphatic hydrogen atoms to standard geometry using the HFIX option improved the value of *R*1 to 0.15. Next, the structure was refined without any geometrical constraints on the non-hydrogen atoms, with anisotropic thermal displacement factors with rigid bond restraints (RIGU option[27]) giving $R1 = 0.118$. However, the bond distances of C2—N2 and O5—C1 were about 10% larger than their standard values in the resultant structure. Therefore, the distance (DFIX) and angle (DANG) restraints were reintroduced with standard deviations of 0.02 Å and 0.04 Å, respectively, and this gave results of $R1 = 0.129$ for 101 $F_o > 4\sigma(F_o)$ intensities, and $R1 = 0.170$ for all 155 intensities at 1.17 Å resolution. The amine and hydroxyl hydrogen atoms were not included in the refinement model as their positions cannot be reliably determined by fiber X-ray diffraction. The geometry of this final model was checked using PLATON[28] software and no geometric anomaly was found. The final structure and the molecular geometry with anisotropic thermal displacement factors are shown in Figure 3. CIF file containing the atomic coordinates and geometrical parameters are provided in the Supporting Information.

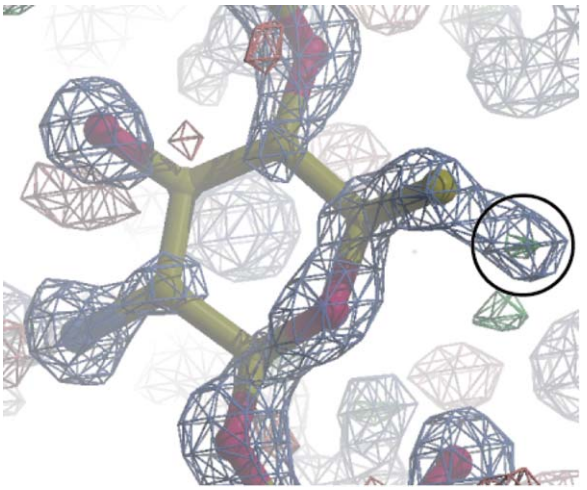


FIGURE 2 Section through the $2mF_o - F_c$ (σ) (showing positive density in blue mesh) and the $F_o - F_c$ (showing positive and negative density in green and red mesh, respectively) Fourier omit map of anhydrous chitosan calculated with the O6 atom omitted. The positive density indicated by the circle can be associated with the omitted O6.

RESULTS AND DISCUSSION

Molecular Arrangement

The crystal structure model of anhydrous chitosan is shown in Figure 3. The molecular geometry and arrangements of previous models are all similar. The Yui model[12] and Okuyama model[13] differ in the orientation of the molecule, rotated by about 90° around the chain axis. Thus the model is almost identical if the coordinates are converted as $(1/4-y,\ x-1/4,\ z+1/4)$. In fact, the *a*- and *b*-axes are only different by 3–4% and the $(h\ k\ l)$ and $(k\ h\ l)$ reflections were treated as overlapping reflections of one spot, making it almost impossible to distinguish the two models based on the X-ray intensity alone. In the Okuyama model, some of the higher order reflections were resolved between $(h\ k\ l)$ and $(k\ h\ l)$, but it is not clear whether those intensities played a decisive role in discriminating the two orientations, as the non-bonded interactions were

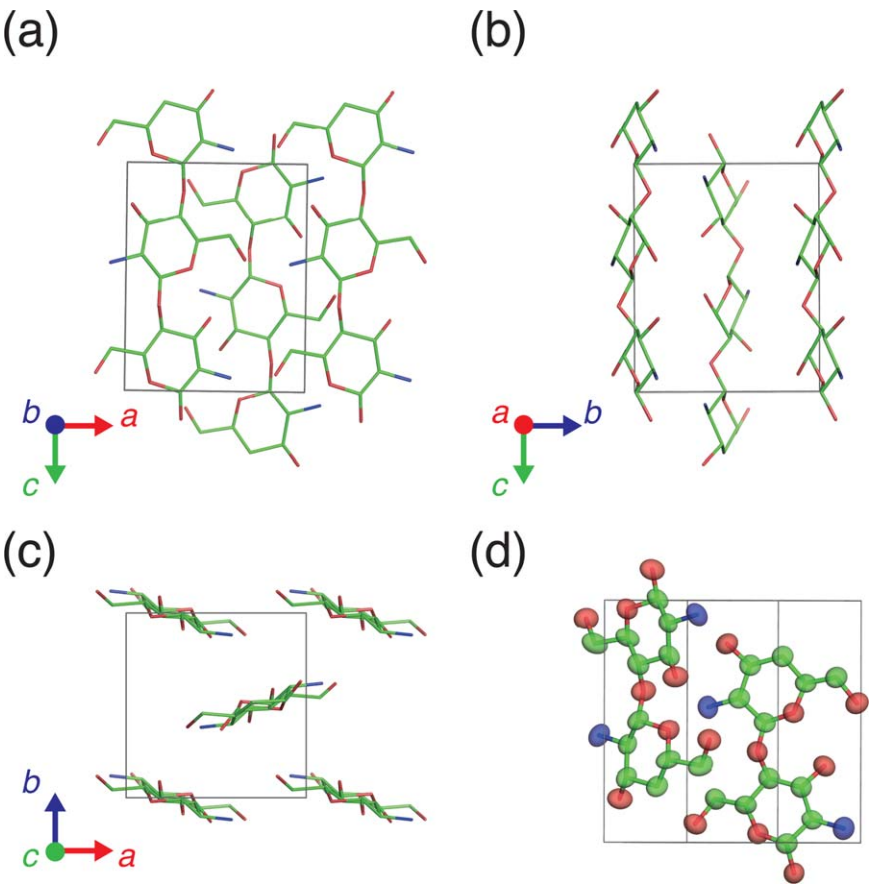


FIGURE 3 Crystal structure of anhydrous chitosan. (a) *ac*-projection, (b) *bc*-projection, (c) *ab*-projection. (d) ORTEP representation. The carbon, nitrogen, and oxygen atoms are represented in green, blue, and red, respectively.

taken into account in the refinement. In our case, the peak positions were well defined, even though we merged the overlapping reflections in the final stage of refinement. For example, one of the strongest peak is centered at the position of 1 2 0 instead of 2 1 0, and the neighboring peaks is 0 2 0 rather than 2 0 0 (Figure 1) making it clear that the pyranose plane should be roughly parallel to the *ac* plane. The Mazeau model[11] is shifted by about 0.75 Å along the chain direction with respect to the other three models including ours. This is probably due to the poor reliability of the non-base-plane data used in the refinement.

The unit cell volume of our proposed model is smaller than the cell parameters previously reported (Table I). The increased volume, as determined by electron diffraction data, is probably due to the electron damage causing swelling along the direction perpendicular to the pyranose plane, as in the case of cellulose.[29] Okuyama et al. and Yui et al. reported similar unit cell parameters, both of which are about 1% larger in all directions. Yui et al. prepared the anhydrous chitosan sample by 240°C hydrothermal annealing, but the annealing duration was not reported. A small amount of residual water might have slightly distorted the lattice while keeping the rough structure. Indeed, the data by Yui et al. does not have a resolution better than 2 Å, and the temperature factor reported by Okuyama et al. is relatively high (12.9 $Å^2$), while that of Yui et al., 0.1 $Å^2$, is exceptionally low. In our case, the diffraction data had atomic resolution, and the global temperature factor was 8 $Å^2$. This is due to the higher perfection of the crystallites that resulted from the annealing method.[8]

The overall density of the crystal determined in this study is lower than those of cellulose allomorphs, but there is a striking similarity in the molecular arrangement, if one focuses on the sheet made of the parallel chains. Figure 4 compares the chitosan structure along the *ac* plane with the corner chains of cellulose II structure[16] along the *ac* plane and the cellulose III_I structure[18] in the *bc* plane. The Cartesian coordinates of cellulose II corner chains and cellulose III_I were converted as $(1.202 + x, -y, 2-z)$ and $(10-cx-sy, sx-cy, z+1.36)$, respectively, where $c = \cos(\gamma = 105°)$ and $s = \sin(\gamma)$. The root mean square deviations (RMSD) of atomic coordinates were

Table I Comparison With Previous Analyses

Method	This study Synchrotron X-ray diffraction	Okuyama et al.[13] Lab source X-ray diffraction	Mazeau et al.[11] Electron diffraction
Sample	Annealed crab tendon chitosan	Via acetic acid complex of crab tendon chitosan	Recrystallized chitosan (DP = 35)
Unit cell parameters			
a (Å)	8.129	8.26	8.07
b (Å)	8.347	8.5	8.44
c (Å)	10.311	10.43	10.34
Number of diffraction spots	101	41	23
R factor	0.127	0.15	0.133
Intramolecular H-bond			
O3–H. . .O5 (Å)	2.73	2.75	2.6
O3–H. . .O6 (Å)	3.08	3.23	3.2
Intermolecular H-bond			
O6. . .N (Å)	2.87	2.72	2.6
N. . .O5 (Å)	3.23	3.67	3.5

estimated between these crystal structures using two pyranose residues in adjacent chains. All the RMSD values were about 0.3 Å (0.31 for chitosan-cellulose II, 0.33 for chitosan-cellulose III_I, and 0.31 for cellulose II-III_I), confirming the structural similarity of these molecular sheet structures.

Looking at the corner chains of cellulose II in the *ac* plane and in the *bc* plane of III_I, structures determined by neutron diffraction, the O2 donates to O6 in the intermolecular hydrogen bond. However, this may not be possible in the case of anhydrous chitosan as the amine is only a weak hydrogen bond donor. A recent DFT and MD calculation of cellulose II and III_I suggests an alternative hydrogen bonding pattern, in which the O6 donates to the O2 in the same configuration.[30] This hydrogen bond pattern is compatible with the anhydrous chitosan structure. The difference in the capacity to donate hydrogen bonding and the new hydrogen bond model would explain the difference in the structure between cellulose II and chitosan. In cellulose II, O2 of the corner chains donates to the center chain pulling it closer to the corner chain, aligning the pyranose plane and making the center chain take an arrangement different from the corner chain. On the other hand, due to the lack of strong interaction between the hydrogen-bonded layers of chitosan, the center chain can take the same structure as the corner chain leading to $P2_12_12_1$ symmetry. The theoretical crystal density of anhydrous chitosan, 1.53 g/cm^3, obtained here, is indeed slightly lower than that of cellulose II (1.58 g/cm^3), but the same as that of cellulose III_I. Elucidating the exact role of hydrogen bonding in explaining the difference in the chain arrangements of cellulose II and anhydrous chitosan will require further studies using neutron diffraction to directly determine the positions of the participating hydrogen atoms.

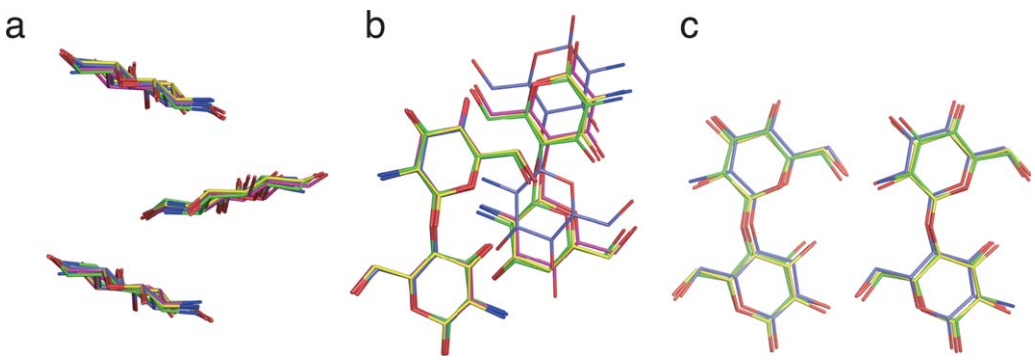


FIGURE 4 Superimposition of crystal structure models of anhydrous chitosan (green, this study: yellow, Okuyama et al.: magenta, Yui et al.: purple, Mazeau et al.) in (a) *ab* projection and (b) *ac* projection. (c) Comparison between sheet structures of chitosan (green), cellulose II (origin layer, yellow), and cellulose III_I (purple).

Table II The Major Conformational Parameters of Anhydrous Chitosan and Other Polysaccharide Crystals

	τ (°)	φ (°)	ψ (°)	χ (°)	χ' (°)	θ (°)
Anhydrous chitosan						
This structure	114	−94	−149	58	180	6.0
Okuyama et al.[13]	117	−96	−145	49	170	3.5
Mazeau et al.[11]	118	−93	−149	48	173	6.9
Other polysaccharides						
α-Chitin	114	−80	−167	47/−44[a]	168/77[a]	16.4
Anhydrous β-chitin[19]	117	−89	−152	−65	58	10.0
Cellulose Iα *up*[14]	116	−99	−140	166	−74	6.9
Cellulose Iα *down*[14]	116	−98	−138	167	−75	9.4
Cellulose Iβ *origin*[15]	115	−99	−142	170	−70	10.2
Cellulose Iβ *center*[15]	116	−89	−147	158	−83	6.7
Cellulose II *origin*[16]	115	−97	−144	72	−165	5.0
Cellulose II *center*[16]	116	−94	−151	58	−175	10.2
Cellulose III[18]	116	−92	−152	44	163	10.5

Glycosidic bond angle τ (C1-O4-C4) and the glycosidic torsion angles φ(O5-C1-O1-C4) and ψ(C1-O1-C4-C5) describe the relative orientation of adjacent glucosyl residues in the same chain. The conformation of the primary hydroxyl group is defined by two torsion angles; χ(O5-C5-C6-O6) and χ'(C4-C5-C6-O6). θ is the Cremer & Pople puckering parameter.

[a] Disordered structure.

Molecular Geometry

The major conformational parameters of anhydrous chitosan and other anhydrous polysaccharide structures are listed in Table II. The τ, φ, and ψ describing the glycosidic bond conformation of anhydrous chitosan have similar value compared to other polysaccharide crystals. The Cremer & Pople puckering parameter θ of this structure is 6°, close to the origin chains of cellulose II, indicating a relaxed pyranose ring.

Table III The Geometry of the Potential Hydrogen Bonds of Anhydrous Chitosan

Donor	Acceptor	Do. . .A (Å)	D. . .A (Å)	Do-D. . .A (°)	A residue	
Strong HB						
Intramolecular						
O3	O5	2.719	—	—	1/2−*x*, −*y*, −1/2+*z*	
O3	O6	3.091	—	—	1/2−*x*, −*y*, −1/2+*z*	
O3	O1	2.979	—	—	*x*, *y*, *z*	
O6	O3	3.091	—	—	1/2−*x*, −*y*, 1/2+*z*	
Intermolecular						
O6	N	2.854	—	—	1+*x*, *y*, *z*	
O3	O6	3.471	—	—	1−*x*, −1/2+*y*, 1/2−*z*	
O6	O3	3.471	—	—	1−*x*, 1/2+*y*, 1/2−*z*	
Weak HB						
Intermolecular						
N	O6	2.854	—	—	−1+*x*, *y*, *z*	
N	O1	3.171	—	—	−1/2+*x*, 1/2−*y*, 1−*z*	
N	O5	3.210	—	—	−1/2+*x*, 1/2−*y*, 1−*z*	
C1H	O6	3.375	2.448	158.3	−1/2+*x*, −1/2−*y*, 1−*z*	A*
C1H	O3	4.129	3.348	137.0	−*x*, 1/2−*y*, 1−*z*	B*
C2H	N	3.894	3.168	132.7	1/2+*x*, 1/2−*y*, 1−*z*	C*
C3H	O3	3.574	2.621	165.1	−*x*, −1/2+*y*, 1/2−*z*	D*
C3H	N	4.051	3.302	134.7	−*x*, −1/2+*y*, 1/2−*z*	E*
C4H	O6	3.471	2.582	151.7	1−*x*, 1/2+*y*, 1/2−*z*	F*
C5H	O6	4.116	3.394	132.8	−1/2+*x*, −1/2−*y*, 1−*z*	G*
C6HA	N	3.938	3.064	150.6	1+*x*, *y*, *z*	H*
C6HB	O1	3.961	3.023	162.9	1/2+*x*, 1/2−*y*, 1−*z*	I*

Do: Donor, D: Donor hydrogen, A: Acceptor, * Labels in Figure 5.

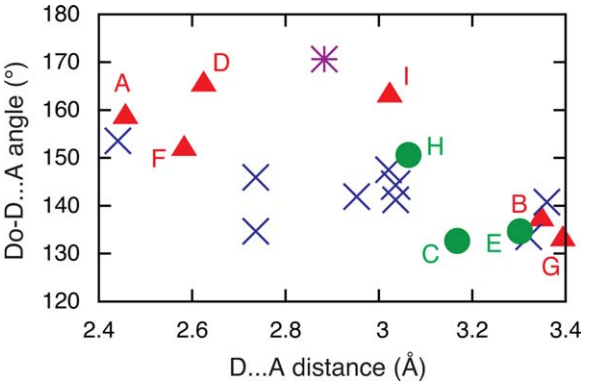


FIGURE 5 Distance–angle distribution of week hydrogen bonds and short contacts. Labels of data points correspond to those in Table III.

The torsion angle χ(O5-C5-C6-O6) of 58° indicates the *gt* conformation of the hydroxymethyl group. This agrees with the fact that the reported ^{13}C NMR chemical shift of C6 of anhydrous chitosan is about 2 ppm higher than the β-chitin which is known to take the *gg* conformation: 61.9 ppm instead of 59.8 ppm.[31] Compared to the non-aminated sugars,[32] the C6 chemical shift of glucosamine seems to be shifted by 2 ppm toward higher magnetic field, keeping the dependency on the hydroxymethyl conformation.

Hydrogen Bonding and Short Contacts

The X-ray data itself does not allow the localization of mobile hydroxyl and amine hydrogen atoms. In addition to the above mentioned O6. . .N hydrogen bonding, the bifurcated intramolecular hydrogen bonding from O3 to O6 and O5 of the neighbor residue is expected, as in the case of cellulose II and III_I, as the overall disposition of the non-hydrogen atoms is the same. The donor-to-acceptor distances are summarized in Table III.

Aliphatic hydrogen bonding can also be involved in weak hydrogen bonding as well as repulsive short contacts. Due to their low degree of freedom, the position of these hydrogen atoms can be reasonably located and the potential hydrogen bonding geometry is summarized in Table III. Figure 5 compares the non-bonded distance of hydrogen to different atoms below 3.4 Å, and the angle of CH. . .A above 120°. Oxygen and nitrogen atoms are potential hydrogen bond acceptor candidates, while the short contact with a hydrogen or carbon atom would not provide electrostatic stabilization and might even cause destabilization. There is a general tendency that the CH. . .O or CH. . .N forms more linear geometry than CH. . .H at a similar distance. The distance between the hydrogen and acceptor oxygen or nitrogen atom correlates well with the angle of the donor-hydrogen-acceptor with the shortest hydrogen bonds being more linear (Figure 5). For example, C1H and C4H point to O6, and C3H points to O3; and one of the C6H points to O1 of the antiparallel neighbor, while the other points to O1 of the same molecule. In fact, all of the aliphatic hydrogen atoms can be considered to be involved in weak hydrogen bonds, stabilizing cohesion between the antiparallel sheets.

The distances between nitrogen atom of the amine group and the glycosidic oxygen O1 or ring oxygen O5 are both around 3.2 Å and suggest the formation of weak hydrogen bond. The positions of the amine and hydroxyl hydrogen atoms are being studied using neutron diffraction to get the full picture of hydrogen bonding and will be reported in a separate paper.

CONCLUSIONS

The crystal structure of anhydrous chitosan was refined to an atomic resolution of 1.17 Å. The position of the pendant O6 atom was determined directly using a Fourier difference map. The molecular packing of the hydrogen-bonded molecular sheet structure is the same as in previous refinements,[11–13] and the overall structure is closest to the structure reported by Okuyama et al.[13] The molecular arrangement in the hydrogen-bonded sheet is very similar to the origin chain of cellulose II along the *ac* plane and cellulose III_I along the *bc* plane. The hydrogen-bonded sheets are packed in an antiparallel fashion, stabilized by weak hydrogen bonding involving all of the aliphatic hydrogens.

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